

Second-dimension outliers (SDO) Model

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This document provides detailed descriptions of all core datasets and codes associated with the study titled “**Second-dimension outliers for spatial prediction.**” It serves as a comprehensive guide to the structure, content, and purpose of each file included in the research package. All codes are fully executable without any modification, provided that the required R packages listed in the scripts are properly installed in the user’s environment. This ensures complete reproducibility of the results presented in the manuscript.

The repository consists of four main folders:

1. **SDO simulation analysis** – containing the codes and data for the simulation experiments that evaluate the performance of the proposed SDO model.
2. **SDO wheat case** – containing the scripts and datasets used in the Australian wheat production case study for spatial prediction and model validation.
3. **SDO validation** – including scripts for sensitivity analysis, robustness testing, and visualization (e.g., Figures 11–14 in the manuscript).
4. **shapefiles** – storing the core shapefile datasets used for spatial mapping and boundary delineation.

All spatial datasets are projected under the GDA 1994 Australia Albers coordinate system to ensure consistency across analyses.

1. SDO_simulation_analysis

dt_sim.rda

The file [*dt_sim.rda*](#) is an R dataset used for the simulation experiment. It contains all necessary data for validating and demonstrating the SDO (Second-Dimension Outlier) model under controlled conditions.

generate_sdo_var.R

The file [*generate_sdo_var.R*](#) contains the core function for generating SDO variables.

This function takes the following input parameters:

- pointlocation: coordinates of the sample points.
- gridlocation: coordinates of the grid cells to be predicted.
- gridvar: variables of the grid cells.
- distuf: buffer distance used for spatial calculation.
- sd: standard deviation value used for identifying outliers.

The function constructs SDO variables by quantifying the spatial outlier characteristics of explanatory variables around unsampled locations. These variables are subsequently used in model training and validation for spatial prediction.

sdo_simdata.R

The file [*sdo_simdata.R*](#) contains the core code for the simulation experiment of the SDO model. This script performs spatial prediction, model validation, and visualization using simulated data.

It is used to generate *Figures 2, 3, and 4* in the manuscript, which illustrate the workflow and performance of the proposed SDO model under controlled simulation conditions.

2. SDO_wheat_case

wheat.rda

The file [*wheat.rda*](#) is an R dataset containing all data used in the wheat production case study. It includes two data frames: dt_points and dt_grids.

- dt_points contains 179 sample points representing Local Government Areas (LGAs). Each record includes the coordinates (x, y) and the corresponding dependent variable wheat production (WP), along with several explanatory variables (AT, ETa, EVI, NDVI, TP, WS, NTO, PTO, SLT, SND).

- [`dt\_grids`](#) represents the prediction grid dataset, containing the coordinates (x, y) of each grid cell, the corresponding predicted WP values, and the same set of explanatory variables as in [`dt\_points`](#).

STE_AUS_shape/STE_Aus_pro.shp

The file [`STE\_AUS\_shape/STE\_Aus\_pro.shp`](#) represents the state boundaries of Australia.

It is used to delineate state-level divisions and serves as the base layer for mapping the spatial distribution of SDO variables, corresponding to [Figure 10](#) in the manuscript.

generate_sdo_var.R

The file [`generate\_sdo\_var.R`](#) has been introduced earlier as the core function for generating SDO variables.

As previously described, it constructs spatial outlier features based on buffer distance and standard deviation parameters, which are later used for model training and validation.

sdo_case.R

The file [`sdo\_case.R`](#) serves as the main script for the Australian wheat production case study. It performs spatial prediction and model validation using the SDO model. Specifically, it generates the initial wheat production distribution map corresponding to [Figure 5](#) (later refined in ArcGIS), produces the correlation analysis plots shown in [Figure 8](#), and visualizes the spatial distributions of SDO variables for AT ([Figure 9](#)) and for TP, NDVI, and SND ([Figure 10](#)). In addition, the script produces the variable importance plots for individual predictors in [Figure 12\(b\)](#) and [Figure 12\(c\)](#), exports the spatial prediction results of the aspatial SVM and SDO-SVM models as [CSV files](#), and computes the performance metrics (R^2 , RMSE, and MAE) for the seven machine learning models reported in [Table 2](#).

predvalue_svm.csv

The file [`predvalue\_svm.csv`](#) contains the spatial prediction results of the SVM model, generated from the script [`sdo\_case.R`](#).

It records the predicted values of wheat production across the Australian wheat belt and serves as a core dataset for visualizing model performance and conducting spatial comparison between the Aspatial SVM and SDO-SVM models.

3. SDO_case_validation

wheat.rda

The file [wheat.rda](#) is the same dataset introduced earlier, containing all data used in the wheat production case study.

generate_sdo_var.R

The file [generate_sdo_var.R](#) has been introduced earlier as the core function for generating SDO variables.

sdo_validation.R

The file [sdo_validation.R](#) is used for validating the SDO model. It generates the cross-section yield comparison plots shown in [Figure 13\(b\)](#) and [Figure 13\(c\)](#) (Sections A and B), as well as the wheat production density distribution plots presented in [Figure 14](#). These visualizations demonstrate the model's performance in capturing spatial variations and extreme values across the Australian wheat belt.

sdo_sensitivity_analysis.R

The file [sdo_sensitivity_analysis.R](#) is used to perform the sensitivity analysis, evaluating the robustness of the proposed SDO model with respect to different parameter settings.

This script systematically tests variations in buffer distance, buffer threshold, and outlier identification threshold (standard deviation) to assess their influence on model performance.

The results of this analysis are visualized in [Figure 11](#) of the manuscript.

CSV files

The CSV files contain spatial prediction results generated by the Aspatial SVM and SDO-SVM models.

- [predvalue_svm.csv](#): Prediction results for the entire Australian wheat belt.
- [predvalue_svm_NSW.csv](#), [predvalue_svm_VIC.csv](#), [predvalue_svm_QLD.csv](#), [predvalue_svm_SA.csv](#), and [predvalue_svm_WA.csv](#): Prediction results for the individual states of New South Wales (NSW), Victoria (VIC), Queensland (QLD), South Australia (SA), and Western Australia (WA), respectively.

These CSV files are used in [sdo_validation.R](#) to generate [Figure 14](#) in the manuscript.

The files [cross_sectionA_points.csv](#) and [cross_sectionB_points.csv](#) contain prediction results along cross sections A and B, respectively, and are used in [sdo_validation.R](#) to produce [Figure 13\(b\)](#) and [Figure 13\(c\)](#).

Figures [13\(a\)](#), [13\(d\)](#), [13\(e\)](#), and [13\(f\)](#) were generated using ArcGIS software for more precise cartographic representation.

4. shapefiles

The shapefiles folder contains the core shapefile (.shp) datasets used in this study.

All shapefiles are projected under the *GDA 1994 Australia Albers* coordinate reference system, ensuring spatial consistency across all analyses.

AUS_Simple_boundary.shp

The file [AUS_Simple_boundary.shp](#) represents the simplified national boundary of Australia.

AUS_State_boundary.shp

The file [AUS_State_boundary.shp](#) delineates the administrative boundaries of all Australian states and territories.

AUS_Wheatbelt_boundary.shp

The file [AUS_Wheatbelt_boundary.shp](#) defines the spatial extent of the Australian wheat belt region, where wheat cultivation is concentrated. This boundary was used to spatially constrain the study area, ensuring that all analyses and model predictions were restricted to the major wheat-producing zones.

Australia_wheat_production_LGA_2021.shp

The file [Australia_wheat_production_LGA_2021.shp](#) contains the 2021 Local Government Area (LGA) boundaries across Australia, along with the corresponding wheat production data for each LGA. It provides the spatial unit of analysis for the wheat production case study and serves as the primary response dataset for model training and validation.

SVM_Prediction_results.shp

The file [SVM_Prediction_results.shp](#) stores the spatial prediction results generated by the SDO-SVM and Aspatial SVM models.

Each record represents a grid cell within the Australian wheat belt, containing predicted wheat production values and related attributes.

cross_sectionA.shp

The file [cross_sectionA.shp](#) represents Cross-section A used in the wheat belt region of New South Wales for spatial comparison between model predictions. This shapefile defines a transect running southwest to northeast, used to extract and visualize the predicted wheat production values from both the SDO-SVM and aspatial SVM models along the same spatial path.

It corresponds to the purple line shown in *Figure 13(a), (d), (e), and (f)* and the trend line presented in *Figure 13(b)*.

cross_sectionB.shp

The file [cross_sectionB.shp](#) represents Cross-section B in the wheat belt region of New South Wales. This shapefile defines a transect running southwest to northeast, used to extract and visualize the predicted wheat production values from both the SDO-SVM and aspatial SVM models along the same spatial path.

It corresponds to the blue line shown in *Figure 13(a), (d), (e), and (f)* and the curve displayed in *Figure 13(c)*.